

Assignment - 26

1) What is diff b/w inheritance in C++ & Java?

→ Basic diff:

C++ - Multiple Inheritance (A class can inherit from 2 or more base classes)

Java - Supports single inheritance.

Multiple Inheritance - Not Allowed
→ Can use interfaces

2) What are different types of inheritance in Java?

→ All are allowed EXCEPT - Multiple.

e.g. Single, multilevel, Hierarchical.

3) Concept of keyword in Java.

→ Current instance of class used to differentiate instance variables from local variables when they have the same name, can be used to invoke current class constructor or method.

4) Concept - Super keyword.

→ Parent class object used to access parent class variable, methods & constructors. Helps in method overriding by calling the superclass version of method.

5) How we can call one constructor from another constructor of same class?

→ Using this() keyword to call another constructor within same class.

```
public class Demo {
```

```
    Demo() {
```

```
        this();
```

```
        sout("Default constructor");
```

```
    }
```


✓ Demo (inse) ?

```
{  
    cout << "Parameterized Constructor" << x << endl;  
}
```

Q1] How we can call constructor from constructor of derived class ?

→ Use of super() to call base class constructor from derived class constructor

Class Base {

```
    Base() { cout << "Base Constructor";  
}
```

}

Class Derived extends Base {

```
    Derived() {
```

```
        super();
```

```
        cout << "Derived Constructor";  
    }
```

}

}

Q2] Again, scenario in which we have to call the base class constructor from derived class explicitly ?

→ When the base class constructor requires parameter it must be explicitly called using super() in derived class constructor.

eg. Class Base {

```
    Base(int x) { cout << "Base Con with value: " << x << endl;  
}
```

}

}

Class Derived extends Base {

```
    Derived() {
```

```
        super(10);
```

```
        cout << "Derived Constructor";  
    }
```

Q1 Write a Java program which accept one word from user through command line & display length of that word.

```
→ public class wordlength {  
    public static void main (String[] args) {  
        if (args.length > 0) {  
            String word = args [0];  
            System.out ("Length is: " + word.length());  
        } else {  
            System.out ("No word entered");  
        }  
    }  
}
```

Q2 Java program which accepts N words from user through command line and display length of largest word.

```
→ public class LargestWordLength {  
    public static void main (String[] args) {  
        int max length = 0;  
        String longestword = " ";  
        for (String word : args) {  
            if (word.length() > max length) {  
                max length = word.length();  
                longestword = word;  
            }  
        }  
        System.out ("Length of largest word is: " + max length);  
    }  
}
```


- 10/ What is use of classpath for java programs?
- It is a command line action that tells the Java Compiler & JVM where to find compiled classes & packages.
 - It is used to specify location of external libraries & compiled class files.